

	Assessment:	Project Milestone 2
	Subject:	Business Intelligence 381
	Total:	50

Project Milestone 2: Data Preprocessing

Socioeconomic Development & Household Well-being in South Africa (SDHSA)

Purpose

Milestone 2 addresses the **Data Preparation** phase of CRISP-DM. Using the evidence established in Milestone 1, your team must design, implement and validate a reproducible preprocessing pipeline that converts the selected source data into a technically sound analytical dataset suitable for the chosen modelling and business intelligence tasks.

Key principle

Data preparation is not only cleaning. It includes selecting relevant records and attributes, resolving grain, integrating datasets correctly, engineering useful features and producing a reproducible pipeline.

Activities in this phase enable you to become familiar with the data, identify data quality problems, discover initial insights, and detect interesting subsets to form hypotheses about hidden patterns or relationships.

Tasks

This milestone consists of the following steps that should be submitted as an assignment on given due date.

Task 1 - Confirm Data Selection

Review Milestone 1 findings and confirm which datasets, records and variables will be retained and used for analysis. Justify inclusion and exclusion decisions in relation to the business and modelling goals.

- Based on data selection criteria, decide if one or more attributes are more important than others and weigh the attributes accordingly. Decide, based on the context (i.e., application, tool, etc.), how to handle the weighting.
- Select relevant data subsets (e.g., significant attributes, and only data which meet certain conditions or use other more advanced techniques). Model requirements may show that other datasets are needed.
- Document the analytical grain of the intended final dataset before joining data.

Task 2 - Clean and Recode

- Using the Milestone 1 quality assessment, identify and justify the treatment of relevant completeness, special-value and coding concerns. Do not apply a cleaning operation merely because it is commonly used.

	Assessment:	Project Milestone 2
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- Assess uniqueness and potential duplication using the correct observation grain and substantive keys. Distinguish genuine duplicate records from legitimate repeated identifiers in one-to-many data.
- Investigate validity, suspicious observations and potential outliers using domain and distributional evidence. Do not automatically delete legitimate extreme observations.
- Recode categorical variables into analytically meaningful forms while retaining traceability to source values.
- Correct data types and standardise inconsistent representations where required, while documenting the original and resulting values or categories.

Task 3 -Integrate and Construct

- Join only datasets that can legitimately be integrated at the chosen grain.
- Demonstrate join cardinality and verify row counts before and after each join.
- Where row-level integration is not valid, use a justified alternative such as aggregation, parallel analysis or evidence synthesis.
- Create derived features that add analytical value and explain their construction.
- Produce a final data dictionary describing retained and engineered variables.

Task 4 -Prepare for Modelling

- Define the modelling target or segmentation feature set.
- Identify and remove identifiers, leakage variables or unsuitable predictors.
- Apply transformations, scaling, encoding or discretisation only where required by the chosen technique.
- Assess class imbalance where applicable.
- Design an appropriate train/test split or resampling strategy and justify it. Prevent preprocessing leakage.
- Produce a final analytical dataset and a reproducible preparation pipeline. Provide before-and-after validation evidence for major preprocessing decisions, including relevant row counts, variable types, category levels, missingness, uniqueness, join cardinality and unmatched records where applicable.

Required Deliverables

- Milestone 2 Data Preparation Report (PDF).
- Reproducible R Markdown/Quarto preparation pipeline with code, output and verification checks.
- Clean/integrated analytical dataset(s) used for Milestone 3.
- Team data dictionary including source and engineered variables.
- Updated ethics and responsible AI usage log.

	Assessment:	Project Milestone 2
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Assessment Rubric

Criterion	Evidence expected	Marks
Data selection & analytical grain	Clear inclusion/exclusion decisions and correct target grain.	7
Cleaning & quality treatment	Evidence-based handling of missing/special values, duplicates, types and outliers.	10
Dataset integration & validation	Correct keys/cardinality, verified joins and defensible handling of incompatible datasets.	12
Feature engineering & transformation	Useful, justified features and appropriate recoding/transformation.	8
Data dictionary & reproducibility	Traceable variables and a pipeline that can be rerun.	5
Modelling preparation & leakage control	Appropriate target/features, train-test/resampling design and leakage prevention.	8
TOTAL:		50

IMPORTANT:

Marks are awarded for demonstrated implementation and validation - not just narrative description. For every major preprocessing decision, clearly document:

- **Problem:** What data quality issue or inconsistency was identified?
- **Treatment:** What method or transformation was applied in R to resolve it?
- **Effect:** How did the dataset change after the transformation?
- **Verification:** What code, summary output, or visualization confirms the fix worked as intended?

Note: Unsupported claims without accompanying R code or output will receive zero or limited marks. Take particular care with relational operations: invalid table joins that duplicate, unintentionally discard, or misalign records will severely impact your work.

Additional Information

- This is a group project. Please continue with your allocated project groups.
- All work must be original. Copying another group's work will not be tolerated.
- Includes names of all group members on the cover page.
- Submit your project milestones electronically on Moodle (BC Connect) before the due dates.
- All writing must be correctly cited and referenced.

	Assessment:	Project Milestone 2
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- **Plagiarism is a serious offence.** Belgium Campus makes use of software that can scan for plagiarism and penalties apply for any plagiarised work including getting a **0** for the assignment.
- No mark will be awarded if the assignment is not uploaded via BC Connect.
- Late assignments will not be accepted; missing the deadline is an automatic **0**.